

$$T\vec{x} = T(x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_n\vec{v}_n)$$

$$= x_1(T\vec{v}_1) + x_2(T\vec{v}_2) + \dots + x_n(T\vec{v}_n)$$

$$= (\underbrace{T\vec{v}_1 \quad T\vec{v}_2 \quad \dots \quad T\vec{v}_n}_{\downarrow \text{From the previous remark}}) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$= (\underbrace{\vec{w}_1 \quad \vec{w}_2 \quad \dots \quad \vec{w}_m}_{\text{each entry is the coefficient of } \vec{w}_i \text{ when multiplying out}}) \underbrace{[T]_{B_2}^{B_1}}_{\text{each entry is the coefficient of } \vec{w}_i \text{ when multiplying out}} [\vec{x}]_{B_1}$$

each entry is the coefficient of $\vec{w}_i$ when multiplying out

$$\Rightarrow [T\vec{x}]_{B_2} = [T]_{B_2}^{B_1} [\vec{x}]_{B_1}$$

Definition: Let $T: V \rightarrow W$ and $U: W \rightarrow Z$ be linear transformations.

$\forall \vec{x} \in V$: define the composition $UT: V \rightarrow Z$ by

$$(UT)(\vec{x}) = U(T(\vec{x})).$$

Proposition: The above composition map $UT: V \rightarrow Z$ is a linear transformation.

Proof: $\forall \vec{x}, \vec{y} \in V$ and $c \in \mathbb{R}$ (or $\mathbb{C}$).

$$\begin{aligned} (UT)(c\vec{x} + \vec{y}) &= U(T(c\vec{x} + \vec{y})) \\ &= U(cT(\vec{x}) + T(\vec{y})) \\ &= cU(T(\vec{x})) + U(T(\vec{y})) \\ &= c(UT)(\vec{x}) + (UT)(\vec{y}) \quad \square \end{aligned}$$

Proposition: Let $T: V \rightarrow W$ and $U: W \rightarrow Z$ be linear transformations.

Let $B_1 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$, $B_2 = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_m\}$, and $B_3 = \{\vec{z}_1, \vec{z}_2, \dots, \vec{z}_k\}$ be

ordered bases for V , W , and Z . Then $[UT]_{B_3}^{B_1} = [U]_{B_3}^{B_2} [T]_{B_2}^{B_1}$. // end of Jan 24

Proof: Note by the definition of matrix representation:

$$(\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m) [T]_{B_1}^{B_2} = T(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) \quad ①$$

$$(\vec{z}_1, \vec{z}_2, \dots, \vec{z}_k) [U]_{B_2}^{B_3} = U(\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m) \quad ②$$

$$\begin{aligned} (\vec{z}_1, \vec{z}_2, \dots, \vec{z}_k) [UT]_{B_1}^{B_3} &= UT(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) \\ &= U(\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m) [T]_{B_1}^{B_2} \quad \text{by ①} \\ &= (\vec{z}_1, \vec{z}_2, \dots, \vec{z}_k) [U]_{B_2}^{B_3} [T]_{B_1}^{B_2} \quad \text{by ②} \end{aligned}$$

$$\Rightarrow (\vec{z}_1, \vec{z}_2, \dots, \vec{z}_k) \left([UT]_{B_1}^{B_3} - [U]_{B_2}^{B_3} [T]_{B_1}^{B_2} \right) = (\vec{0} \quad \vec{0} \quad \dots \quad \vec{0})$$

As $\{\vec{z}_1, \dots, \vec{z}_k\}$ are linearly independent.

$$[UT]_{B_1}^{B_3} = [U]_{B_2}^{B_3} [T]_{B_1}^{B_2}$$

Proposition. (change-of-bases). Let $T: V \rightarrow W$ be a linear transformation.

Let $B_1 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ and $B'_1 = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ be ordered bases for V .

$B_2 = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_m\}$ and $B'_2 = \{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m\}$ be ordered bases for W .

Suppose $[\vec{x}_i]_{B_1} = \vec{p}_i \in \mathbb{R}^n$ (or $\mathbb{C}^n$), $\forall i=1, \dots, n$, and denote $P = (\vec{p}_1, \vec{p}_2, \dots, \vec{p}_n) \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$)

Suppose $[\vec{y}_j]_{B_2} = \vec{q}_j \in \mathbb{R}^m$ (or $\mathbb{C}^m$), $\forall j=1, \dots, m$ and denote $Q = (\vec{q}_1, \vec{q}_2, \dots, \vec{q}_m) \in \mathbb{R}^{m \times m}$ (or $\mathbb{C}^{m \times m}$)

Then: $[T]_{B'_1}^{B'_2} = Q^{-1} [T]_{B_1}^{B_2} P$ or equivalently $[T]_{B_1}^{B_2} = Q [T]_{B'_1}^{B'_2} P^{-1}$

Remark: One may prove that P and Q are both invertible matrix.

Proof: As $[\vec{x}_i]_{B_1} = \vec{p}_i \Rightarrow \vec{x}_i = (\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) p_i$ for all $i=1, \dots, n$.

$$\Rightarrow (\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) = (\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) P \quad ①$$

$$\text{similarly } (\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m) = (\vec{w}_1, \vec{w}_2, \dots, \vec{w}_m) Q \quad ②$$

By the definition of matrix representation:

$$T(\vec{v}_1, \dots, \vec{v}_n) = (\vec{w}_1, \dots, \vec{w}_m) [T]_{B_1}^{B_2} \quad ③$$

$$T(\vec{x}_1 \dots \vec{x}_n) = (\vec{y}_1 \dots \vec{y}_m) [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} \quad (4)$$

use (1) and (2) to replace $(\vec{x}_1 \dots \vec{x}_n)$ and $(\vec{y}_1 \dots \vec{y}_m)$ in (4), we get:

$$T(\vec{v}_1 \dots \vec{v}_n)P = (\vec{w}_1 \dots \vec{w}_m) Q [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'}$$

$$\Leftrightarrow T(\vec{v}_1 \dots \vec{v}_n) = (\vec{w}_1 \dots \vec{w}_m) Q [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1} \quad (5)$$

Compare (5) with (3): we got $(\vec{w}_1 \dots \vec{w}_m) Q [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1} = (\vec{w}_1 \dots \vec{w}_m) [T]_{\mathcal{B}_1}^{\mathcal{B}_2}$

Since $\{\vec{w}_1 \dots \vec{w}_m\}$ is linearly independent $Q [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1} = [T]_{\mathcal{B}_1}^{\mathcal{B}_2}$

$$\Leftrightarrow [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} = Q^{-1} [T]_{\mathcal{B}_1}^{\mathcal{B}_2} P.$$

A second proof: As $(\vec{x}_1 \vec{x}_2 \dots \vec{x}_n) = (\vec{v}_1 \vec{v}_2 \dots \vec{v}_n) P$

$$\forall \vec{x} \in V: \quad \vec{x} = (\vec{x}_1 \vec{x}_2 \dots \vec{x}_n) [\vec{x}]_{\mathcal{B}_1'} = (\vec{v}_1 \vec{v}_2 \dots \vec{v}_n) P [\vec{x}]_{\mathcal{B}_1'} \\ = (\vec{v}_1 \vec{v}_2 \dots \vec{v}_n) [\vec{x}]_{\mathcal{B}_1}$$

$$\Rightarrow P [\vec{x}]_{\mathcal{B}_1'} = [\vec{x}]_{\mathcal{B}_1} \Rightarrow [\vec{x}]_{\mathcal{B}_1'} = P^{-1} [\vec{x}]_{\mathcal{B}_1}$$

$$\text{similarly, } \forall \vec{y} \in W: Q [\vec{y}]_{\mathcal{B}_2'} = [\vec{y}]_{\mathcal{B}_2} \Rightarrow [\vec{y}]_{\mathcal{B}_2'} = Q^{-1} [\vec{y}]_{\mathcal{B}_2}$$

$$\Rightarrow \text{As } T\vec{x} \in W: [T\vec{x}]_{\mathcal{B}_2'} = Q^{-1} [T\vec{x}]_{\mathcal{B}_2} = Q^{-1} [T]_{\mathcal{B}_1}^{\mathcal{B}_2} [\vec{x}]_{\mathcal{B}_1}$$

$$\text{On the other hand: } [T\vec{x}]_{\mathcal{B}_2'} = [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} [\vec{x}]_{\mathcal{B}_1'} = [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1} [\vec{x}]_{\mathcal{B}_1}$$

$$\Rightarrow Q^{-1} [T]_{\mathcal{B}_1}^{\mathcal{B}_2} [\vec{x}]_{\mathcal{B}_1} = [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1} [\vec{x}]_{\mathcal{B}_1}$$

$$\Rightarrow Q^{-1} [T]_{\mathcal{B}_1}^{\mathcal{B}_2} = [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} P^{-1}$$

$$\Rightarrow [T]_{\mathcal{B}_1'}^{\mathcal{B}_2'} = Q [T]_{\mathcal{B}_1}^{\mathcal{B}_2} P. \quad \square$$

Definition: let $A, B \in \mathbb{R}^{m \times n}$ or $(\mathbb{C}^{m \times n})$. If there exists invertible matrices

$P \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) and $Q \in \mathbb{R}^{m \times m}$ (or $\mathbb{C}^{m \times m}$) such that $A = QBP$.

Then A and B are called equivalent matrices.

Corollary: Let $T: V \rightarrow V$ be a linear transformation on V .

Let $B = \{\vec{v}_1, \dots, \vec{v}_n\}$ and $B' = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ be ordered bases for V .

Suppose: $[\vec{x}_i]_B = \vec{p}_i \in \mathbb{R}^n$ (or $\mathbb{C}^n$), and denote $P = (\vec{p}_1 \dots \vec{p}_n) \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$).

$$\text{Then } [T]_{B'} = P^{-1} [T]_B P.$$

Definition: Let $A, B \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$). If there exists an invertible matrix $P \in \mathbb{R}^{n \times n}$ (or $\mathbb{C}^{n \times n}$) such that $A = P^{-1} B P$. Then A and B are called similar matrices.

Proposition: Let V and W be finite-dimensional vector spaces. Let $B_1 = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ and $B_2 = \{\vec{y}_1, \vec{y}_2, \dots, \vec{y}_m\}$ be ordered bases for V and W , respectively.

Then the map $\phi: L(V, W) \rightarrow \mathbb{R}^{m \times n}$ (or $\mathbb{C}^{m \times n}$)

$$T \longmapsto [T]_{B_2}^{B_1}$$

is an isomorphism.

Proof: Need to prove: ① ϕ is a linear transformation, and

② ϕ is one-to-one

③ ϕ is onto.

Proof: For ①: $\forall T_1, T_2 \in L(V, W)$, and $c \in \mathbb{R}$ (or $\mathbb{C}$)

$$\text{As for each } x_i: (cT_1 + T_2)(\vec{x}_i) = cT_1(\vec{x}_i) + T_2(\vec{x}_i)$$

$$\Rightarrow [(cT_1 + T_2)(\vec{x}_i)]_{B_2} = [cT_1(\vec{x}_i) + T_2(\vec{x}_i)]_{B_2}$$

$$= c[T_1(\vec{x}_i)]_{B_2} + [T_2(\vec{x}_i)]_{B_2}$$

$$\begin{aligned}
 \phi(cT_1 + T_2) &= [cT_1 + T_2]_{\beta_1}^{\beta_2} \overset{\text{Column by column}}{=} \begin{bmatrix} \downarrow \text{first column} & & \downarrow \text{the } n\text{-th column} \\ c[T_1(x_1)]_{\beta_2} + [T_2(x_1)]_{\beta_2} & \dots & c[T_1(x_n)]_{\beta_2} + [T_2(x_n)]_{\beta_2} \end{bmatrix} \\
 &= c \begin{bmatrix} [T_1(x_1)]_{\beta_2} & \dots & [T_1(x_n)]_{\beta_2} \end{bmatrix} + \begin{bmatrix} [T_2(x_1)]_{\beta_2} & \dots & [T_2(x_n)]_{\beta_2} \end{bmatrix} \\
 &= c[T_1]_{\beta_1}^{\beta_2} + [T_2]_{\beta_1}^{\beta_2} = c\phi(T_1) + \phi(T_2).
 \end{aligned}$$

$$\Rightarrow \phi(cT_1 + T_2) = c\phi(T_1) + \phi(T_2)$$

$$\textcircled{2} \text{ If } \phi(T) = 0 \Rightarrow [T]_{\beta_1}^{\beta_2} = (\vec{0} \dots \vec{0})$$

$$\Rightarrow T = 0 \leftarrow \text{the zero linear transformation}$$

$$\Rightarrow \text{Null}(\phi) = \{0\}$$

$$\Rightarrow \phi \text{ is one-to-one.}$$

$$\textcircled{3} \text{ } \forall A = \mathbb{R}^{m \times n} \text{ (or } \mathbb{C}^{m \times n}). \text{ denote } A = (a_{ij})$$

$$\text{define } T(\vec{x}_i) = (\vec{y}_1 \dots \vec{y}_m) \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{mi} \end{pmatrix}, \text{ for all } i=1, 2, \dots, n.$$

$$\text{And for any } \vec{v}_1, \vec{v}_2 \in V. \text{ and } \forall c \in \mathbb{R} \text{ or } \mathbb{C}:$$

$$T(c\vec{v}_1 + \vec{v}_2) = cT(\vec{v}_1) + T(\vec{v}_2).$$

$$\text{By the definition } T \in \mathcal{L}(V, W), \text{ and } \forall \vec{x} = a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n \in V.$$

$$T(\vec{x}) = a_1 T(\vec{x}_1) + \dots + a_n T(\vec{x}_n)$$

$$= (T(\vec{x}_1) \dots T(\vec{x}_n)) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \quad \square.$$

end of Jun 27